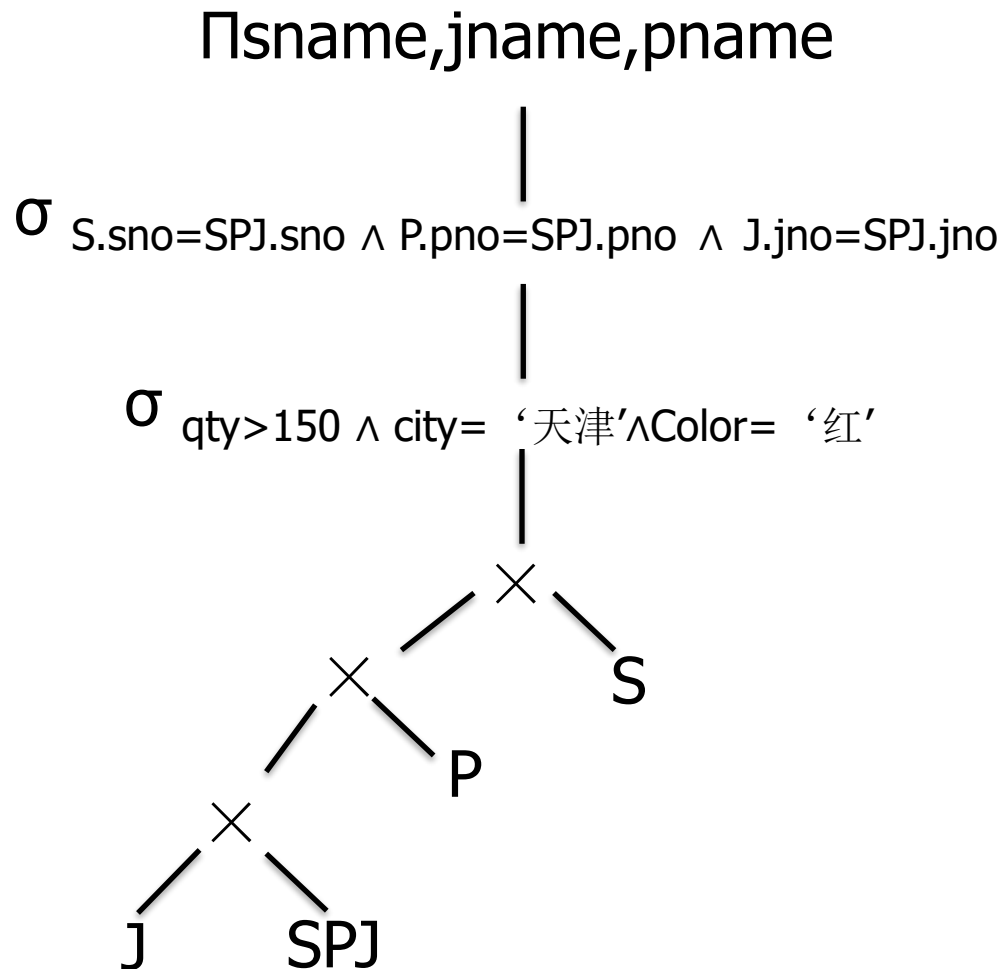


第四次作业

- ```
SELECT sname,jname,pname
FROM S,P,J,SPJ
WHERE S.sno=SPJ.sno
 AND P.pno=SPJ.pno AND J.jno=SPJ.jno
 AND qty>150 AND color='红' AND J.city='天津';
```

请给出此查询语句的关系代数表达式，并画出其关系代数表达的初始查询语法树和优化后的语法树。

$\Pi_{\text{sname}, \text{jname}, \text{pname}} (\sigma_{\text{S.sno}=\text{SPJ.sno} \wedge \text{P.pno}=\text{SPJ.pno} \wedge \text{J.jno}=\text{SPJ.jno} \wedge \text{qty} > 150 \wedge \text{city} = \text{'天津'} \wedge \text{Color} = \text{'红'}} (\text{S} \times \text{SPJ} \times \text{P} \times \text{J}))$



$\Pi_{\text{name}, \text{jname}, \text{pname}}$

$\sigma_{S.\text{sno}=\text{SPJ}.\text{sno}}$

$\sigma_{P.\text{pno}=\text{SPJ}.\text{pno}}$

$\sigma_{J.\text{jno}=\text{SPJ}.\text{jno}}$

$\sigma_{\text{qty}>150}$

$\sigma_{\text{city}=\text{'天津'}}$

$\sigma_{\text{Color}=\text{'红'}}$

